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Department of Philosophy
Columbia University
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Areas of Specialty (AOS): Metaphysics, Logic

Areas of Competence (AOC): Kant, Epistemology, Ethics, Philosophy of Science

Education

Columbia University Ph.D in Philosophy, GPA: 4.11	2019–27 (<i>expected</i>)*
Columbia University B.A. in Mathematics with Philosophy Minor, <i>Magna Cum Laude</i> Honors Thesis: <i>Representing Coxeter Groups</i> (Advisor: Dusa McDuff)	2015–19
University of Cambridge Visiting Student, Oxbridge Scholar in Mathematics	2017–18
Georgia Institute of Technology Calculus III, IV as high school student	2014–15

Dissertation: *Essays on the Antinomies of Space*

Committee: Achille Varzi (primary), Tamar Lando, Michael Thaddeus (Department of Mathematics)

Metaphysical theories of space provide the conceptual foundation for formal theories of spatial representation and reasoning, underpinning modern approaches to computer vision, spatial cognition, and robotics. Since antiquity, however, philosophers have remarked that the metaphysical nature of space itself generates deep and seemingly irresolvable paradoxes. Rather than treating these paradoxes as mere anomalies or obstacles to scientific progress, my dissertation argues that they are indispensable guides to the development of more adequate theories of space and spatial experience. Drawing on tools from geometric analysis, I construct philosophically novel and mathematically rigorous resolutions to the three classic spatial paradoxes: (1) Anaxagoras’s problem of gunk, concerning the infinite divisibility of space; (2) Zeno’s paradox of measure, concerning spatial extension; and (3) Kant’s worries about incongruent counterparts, concerning handedness, or chirality. Across three chapters, I develop a new theory of spatial composition using Fourier analysis, a new account of spatial extension using geometric measure theory, and a new characterization of chirality using projection theory. In each case, I argue that the new formalizations have far-reaching and unexpected metaphysical consequences, including that the well-foundedness of space does not require spatial atoms, that spatial measure cannot be derived from mereological structure alone, and that handedness can be explained within a relationalist ontology of space.

*On official leave of absence for two academic years (2021–2022, 2023–2024).

Publications

- "Generating Gunk," *Journal of Philosophical Logic* 55 (2):377–409 (2026).
- "Modelling Morality: A Kantian Account of Moral Examples," *Kantian Review* 31 (1):71-92 (2026).

Presentations

- "On the Difference Between Left and Right Hands," *BLOC Philosophy of Physics Graduate Student Workshop, University of Oxford* 2026
- "Generating Gunk," *CUNY Logic and Metaphysics Workshop* 2025
- "Introduction to Riemannian Manifolds and Quotient Manifolds," *Brainard Color Vision Lab, University of Pennsylvania* 2025
- "Modelling Morality: A Kantian Account of Moral Examples," *Society for Women in Philosophy (SWIP-NYC)* 2022
- "Models and Values in Science," *Sorbonne-Columbia Blame and Transition Workshop* 2022

Awards & Honors

- Charles Frankel Memorial Fellowship, *GSAS Columbia University* 2024
- Benjamin Wolstein Fellowship, *GSAS Columbia University* 2022
- Graduate Essay Prize (\$500), *Society for Women in Philosophy (SWIP-NYC)* 2022
- Provost Diversity Fellowship (\$8000), *GSAS Columbia University* 2019
- James Gutmann Prize in Philosophy (\$2000), *Columbia University* 2019
- Departmental Honors (Top 10% in Mathematics Major), *Columbia University* 2019
- Oxbridge Scholar (Mathematics), *University of Cambridge* 2017
- Milstein Scholar, *Columbia University* 2015-19

Teaching

*all teaching at Columbia University unless specified otherwise

Sole Instructor

Introduction to Symbolic Logic *Summer 2025*

Teaching Assistant

Epistemology (Jessica Collins) *Spring 2025*

Including One Guest Lecture: "Bullshit & Information Content"

Introduction to Logic (Christopher Prodoehl) *Spring 2023*

Symbolic Logic (Tamar Lando) *Fall 2023 & Fall 2022*

Including One Guest Lecture: "Formal Proofs & Counterexamples in FOL"

Ethics (Michele Moody-Adams) *Spring 2021*

Methods & Problems of Philosophical Thought (Akeel Bilgrami)

Fall 2020

Grader

Advanced Introduction to Bioethics (NYU)

Fall 2024 & Fall 2025

Environmental Ethics (NYU)

Spring 2025

Service

- Referee, *Noûs* 2023
- Colloquium chair, *APA Eastern Division* 2021
- Co-organizer, *SWIP-NYC (Society for Women in Philosophy in New York)* 2020-23
- Graduate Student Representative, *Columbia University* 2022-23
- Co-organizer, *Columbia-NYU Graduate Philosophy Conference* 2019-20
- Research Assistant (Professor: Katja Vogt), *Columbia University* 2020
- Book Assistant (Created index, *Clarke-Doane's Morality and Mathematics*), *OUP* 2019
- Researcher, Thematic Program in Kähler Geometry, *Notre Dame University* 2018
- Researcher, Mathematics REU, *The University of Chicago* 2017

Outreach

- Leader, Differential Geometry for Color Science Working Group, *Brainard Color Vision Lab, University of Pennsylvania* 2024-26
- Volunteer Fellow, *Ruttenberg Cancer Treatment Center, Mount Sinai* 2025-26
- Founder/Writer, *Puzzles to Puzzle You (Remastered) Newsletter* 2025
- Publications Intern, *Artforum International Magazine* 2019
- Publications Intern, *Museum of Modern Art (MoMA)* 2018
- Varsity Athlete, NCAA Division I Women's Tennis, *Columbia University* 2015-7²

Languages

World: French, Telugu (native)

Computer: Python, MATLAB, $\text{\LaTeX}$

²Five-star tennis recruit, ranked top 30 in US. Left team due to injury.

References

Achille Varzi

John Dewey Professor of Philosophy
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Tamar Lando

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Patricia Kitcher

Roberta and William Campbell Professor
Emerita of the Humanities
Columbia University
pk206@columbia.edu

Jessica Collins

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Columbia University
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